

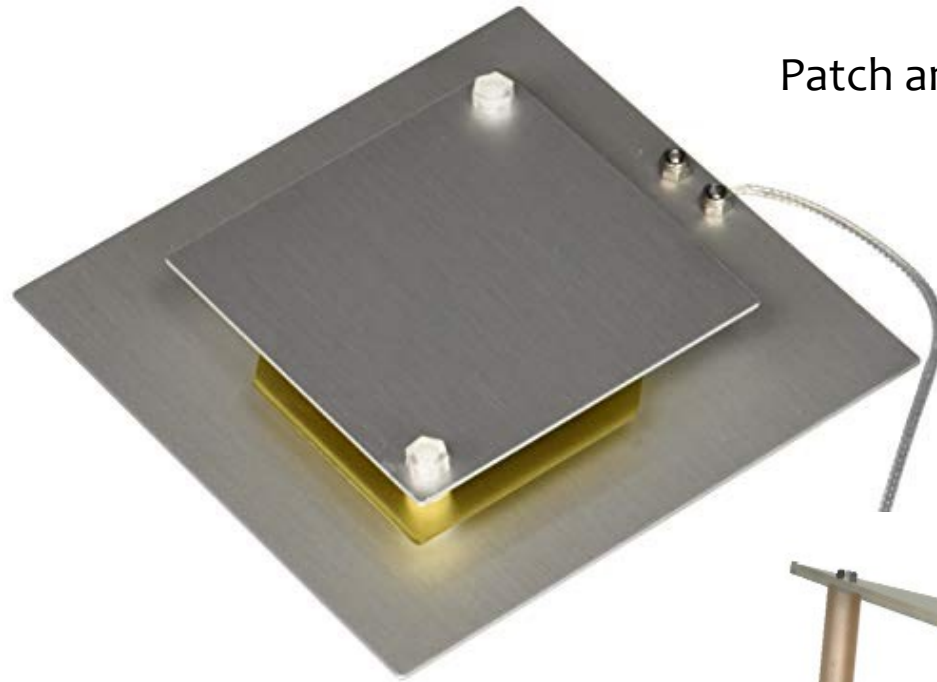
ANTENNAS, INTRODUCTION

Definition of antenna (according to IEEE):

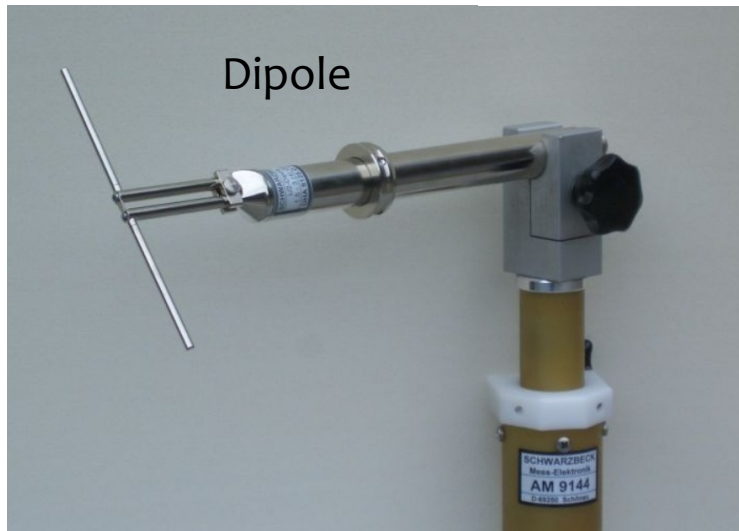
that part of a transmitting or receiving system that is designed to radiate or to receive electromagnetic waves.



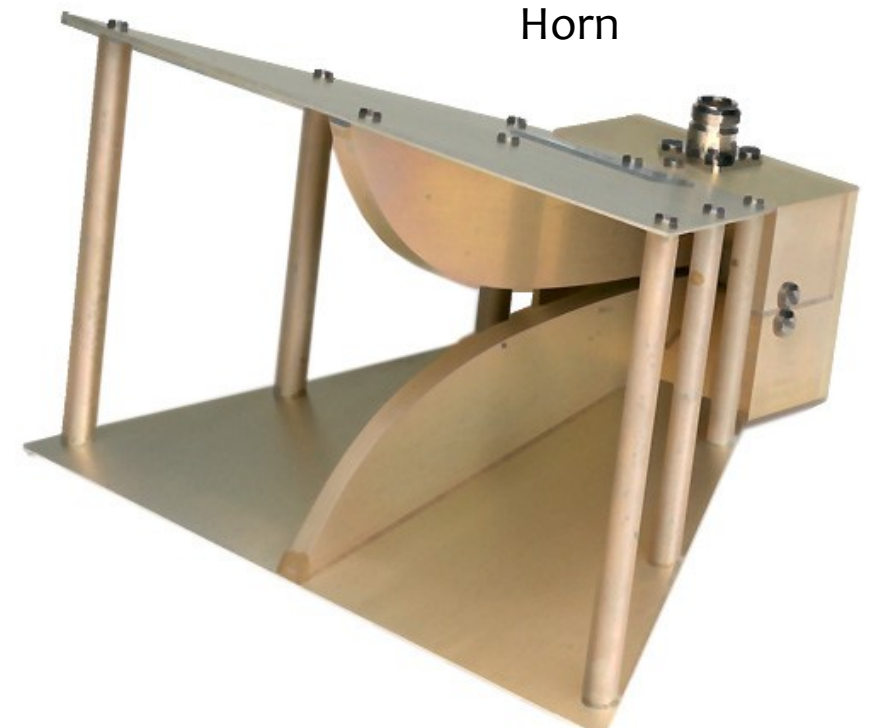
Monopole



Patch antenna



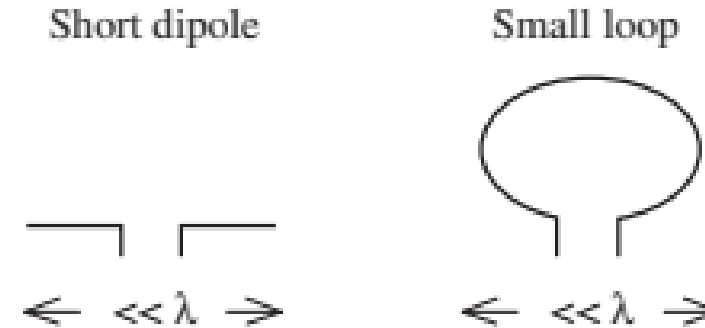
Dipole



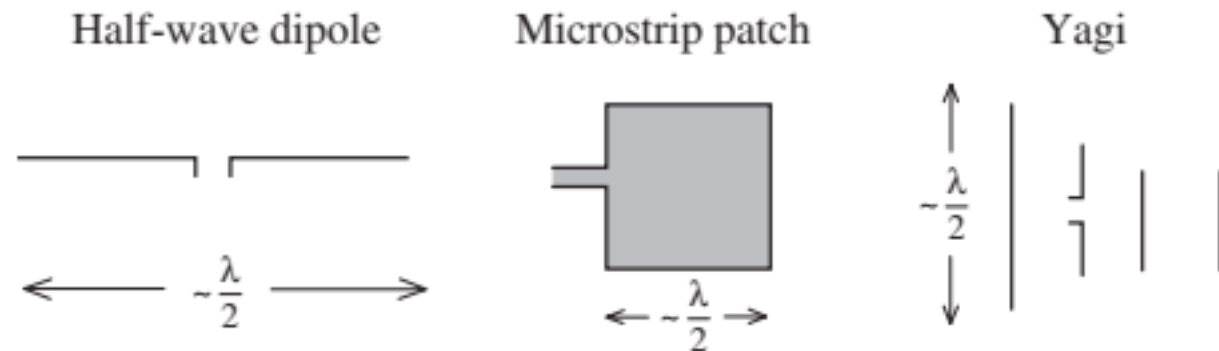
Horn

ANTENNA TYPES

- Electrically small antennas: the extent of the antenna structure is much less than a wavelength
 - Low directivity
 - Low input resistance and high input reactance
 - Low radiation efficiency



- Resonant antennas: these antennas operate well at a single or selected narrow frequency bands
 - Low to moderate gain
 - Real input impedance
 - Narrow bandwidth



ANTENNA TYPES

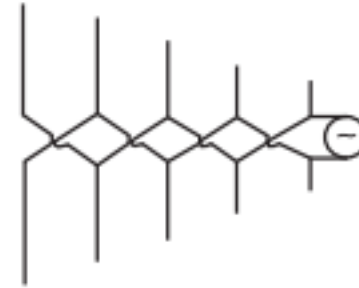
- Broadband antennas: the pattern, gain, and input impedance remain acceptable and are nearly constant over a wide frequency range

- Low to moderate constant gain
- Real input impedance
- Wide bandwidth

Spiral



Log Periodic Dipole Array



- Aperture antennas: these antennas have a physical aperture (or opening) through which electromagnetic waves flow

- High gain, which increases with frequency
- Moderate bandwidth

Horn



Reflector



PRELIMINARY COURSE OUTLINE (the content of the course may be different from that of the previous academic year!)

- Fundamentals of electromagnetics and radiation
- Simple radiating systems (small dipole and small loop)
- Radiation patterns and antenna parameters
- Communication links
- Wire antennas
- Low-profile antennas (microstrip patch)
- Array antennas

ASSESSMENT METHOD

Mandatory written examination comprising theoretical questions and exercises on antenna analysis or design.

REFERENCE MATERIAL

Lesson slides available for download from the E-learning platform



Starting from <https://elearning.unibs.it/> look for [ANTENNAS \(A.A. 2025\)](#) referring to the academic year 2025/26

Recommended bibliography

- Antenna Theory and Design, W.L. Stutzman, G.A. Thiele, 3rd edition, New York, John Wiley & Sons, 2012
- Antenna Theory: Analysis and Design, C.A. Balanis, 4th edition, Hoboken, Wiley, 2016
- Antennas for All Applications, J.D. Kraus, R. Marhefka, 3rd edition, New York, Mc-Graw-Hill, 2002
- Microwave Engineering, D.M. Pozar, 4th edition, Hoboken, Wiley, 2012
- Electromagnetic Waves, C.G. Someda, 2nd edition, Boca Raton, Taylor & Francis, 2006